

Air University



Operating Systems (Lab)

Year 2026

Ms. Fouzia Riaz

Academic Year: 2024 – 2028

Department of Cyber Security

USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

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Lab Task # 04

Part 1: Student Information Program

```
khan_asad@Khan-Asad:~/Desktop$ vi info.cpp
```

```
int main() {
    string name, studentClass;
    int studentID;

    // Asking user to enter information
    cout << "Enter Name: ";
    getline(cin, name);

    cout << "Enter Student ID: ";
    cin >> studentID;

    cout << "Enter Class: ";
    cin >> studentClass;

    // Displaying information in proper format
    cout << "\n---- Student Information ----" << endl;
    cout << "Name: " << name << endl;
    cout << "ID: " << studentID << endl;
    cout << "Class: " << studentClass << endl;

    return 0;
}
```

```
}
:wq!
```

```
khan_asad@Khan-Asad:~/Desktop$ g++ info.cpp -o execute
khan_asad@Khan-Asad:~/Desktop$ ./execute
```

Enter Name: Asad

Enter Student ID: 247171

Enter Class: CYS-A

---- Student Information ----

Name: Asad

ID: 247171

Class: CYS-A

Part 2: Simple Arithmetic Operation (Runtime Input)

```
ibtehaj@ibtehaj: ~/Desktop
#include <iostream>
using namespace std;

int main() {
    double num1, num2;
    cout << "Enter first number: ";
    cin >> num1;
    cout << "Enter second number: ";
    cin >> num2;

    cout << "\nAddition: " << num1 + num2 << endl;
    cout << "Subtraction: " << num1 - num2 << endl;
    cout << "Multiplication: " << num1 * num2 << endl;
    cout << "Division: " << num1 / num2 << endl;

    return 0;
}
```

```
ibtehaj@ibtehaj:~/Desktop$ vi program_a.cpp
ibtehaj@ibtehaj:~/Desktop$ g++ program_a.cpp -o part2
ibtehaj@ibtehaj:~/Desktop$ ./part2

ibtehaj@ibtehaj:~/Desktop$ ./part2
Enter first number: 5
Enter second number: 2

Addition: 7
Subtraction: 3
Multiplication: 10
Division: 2.5
ibtehaj@ibtehaj:~/Desktop$
```

Part 3: Command-Line Argument Concept

```
usama@ubuntu:~/Desktop$ vi program1.cpp
```

```
#include <iostream>
using namespace std;
int main(int argc, char *argv[])
{
    cout << "Program Name: " << argv[0] << endl;
    if(argc == 1)
    {
        cout << "No extra arguments passed." << endl;
    }
    else
    {
        cout << "Number of arguments: " << argc << endl;
        for(int i=0; i<argc; i++)
        {
            cout << "argv[" << i << "]: " << argv[i] << endl;
        }
    }
    return 0;
}

~
~
~
~
~
~
~
~
~
~
:wq!
```

```
usama@ubuntu:~/Desktop$ g++ program1.cpp -o program
usama@ubuntu:~/Desktop$ ./program
Program Name: ./program
No extra arguments passed.
usama@ubuntu:~/Desktop$ ./program Asad Ibtehaj Usama
Program Name: ./program
Number of arguments: 4
argv[0]: ./program
argv[1]: Asad
argv[2]: Ibtehaj
argv[3]: Usama
usama@ubuntu:~/Desktop$
```

THE END
